

Summary: Creating Compelling Reports In Power BI

SESSION OVERVIEW:

By the end of this session, students will be able to:

- Format visuals and pages and create beautifully formatted reports using formatting options.
- Use the drill through feature in Power BI.
- Create bookmarks in Power BI.
- Utilise pre-existing themes and create new themes in Power BI.
- Add other elements to the report to make it more appealing.

KEY TOPICS AND EXAMPLES:

Note: For this session, we will be using [this dataset](#).

1. Formatting visuals and pages

In the visualisation pane, apart from the **Build visual** tab (refer Image 1), there is the **Format visual** tab (refer Image 2). This tab has options to format your visual or the entire page.

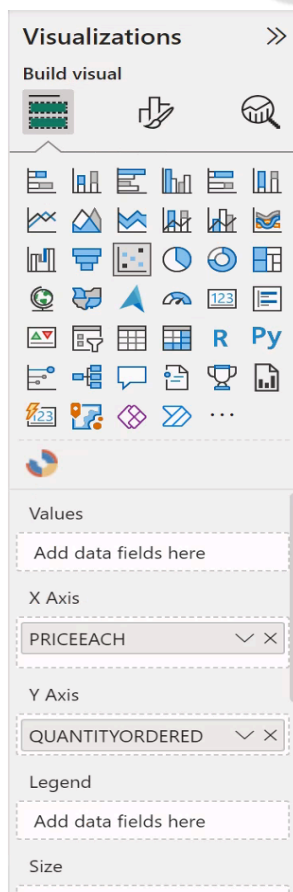


Image 1

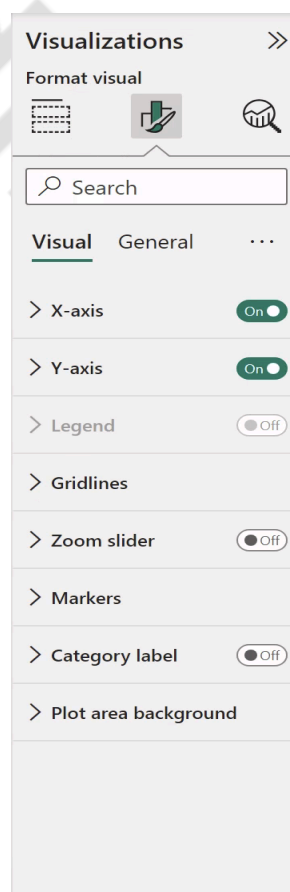


Image 2

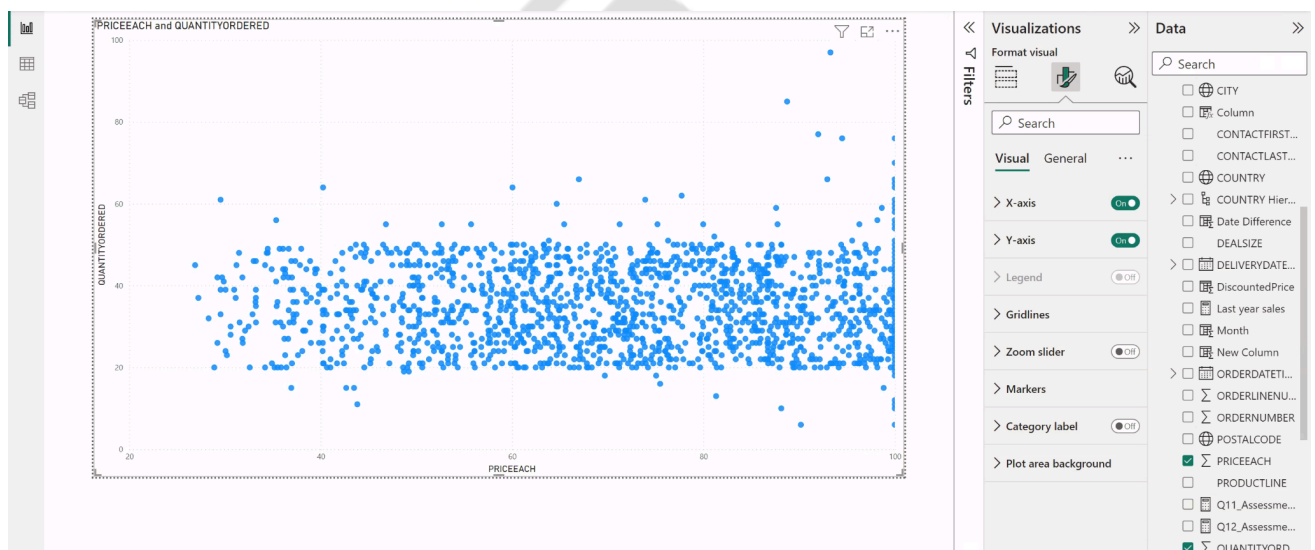
A. Format Visual

The format visual pane has three options:

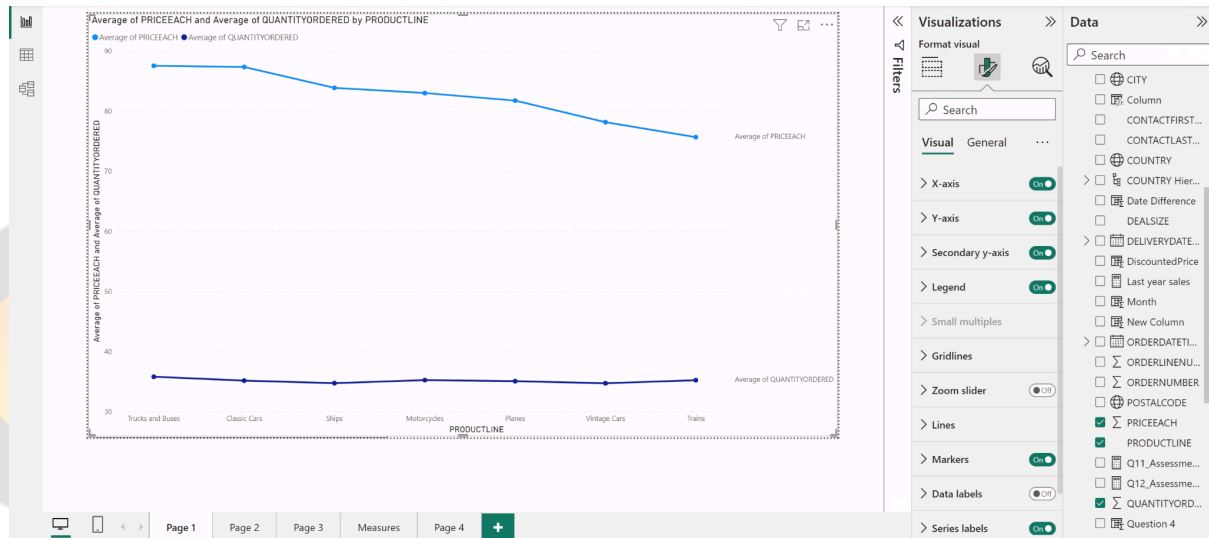
1. Visual

The options vary for each visual. Some of the important options here include:

- **Legend:** If we turn the legend toggle to ON, it will show the legend of the colours. It has multiple sections such as Options (Position), Text (Font, Size, Colour), Title (Font, Size, Colour).
- **Colours:** You can change the colour for each category in your dataset.
- **Data labels:** You can turn this on to show data labels in your chart. If it is turned on it shows a Values section where you can change the Font Style, Size, Colour of the data labels.
- **Category labels:** If this is turned on you can add labels to categories. It has a Values section in which we can change the Font, Size, Colour of the category labels.
- **Map settings:** This appears in the case of a map visual. We can change the Style of the map (Dark, Road, Light, Grayscale, Aerial) and the Controls (Auto-zoom, zoom buttons, lasso button).
- **Bubbles:** This appears in the map visual. We can change the size of the bubbles and the colours for each category.
- **X-axis, y-axis:** These options appear whenever axes are involved. E.g. scatter plot (see image below). We can change the range; the font, colour and size of values; and the Title (title style, title text, font, size, colour etc.).



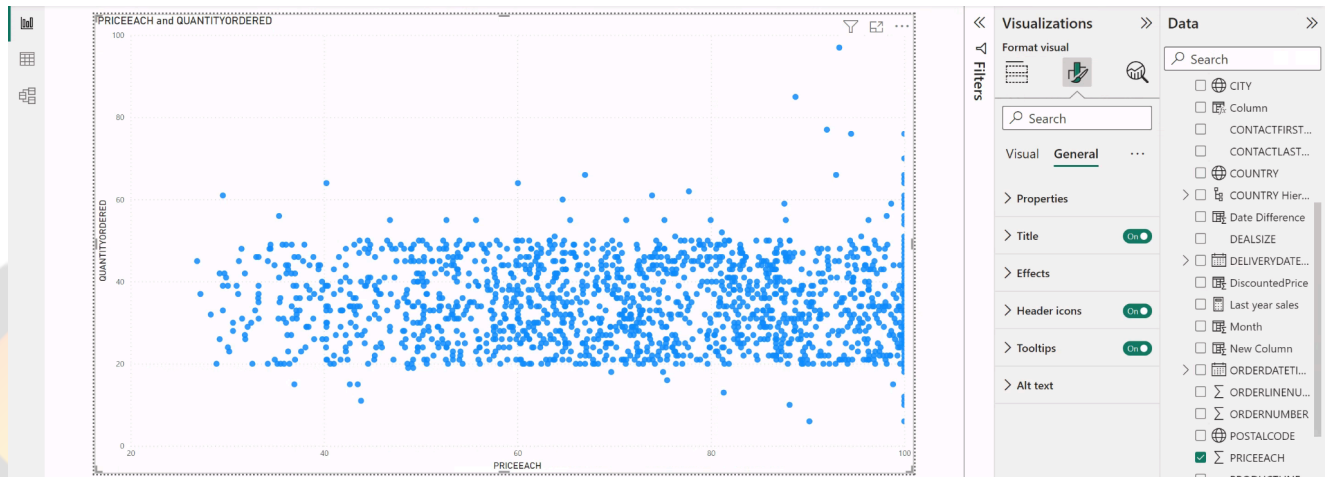
- **Secondary y-axis:** This option comes in case of a line chart, area chart and combination chart. We can change the range, the values (Font, Size, Colour etc) and the title.
- **Gridlines:** It includes options to change the gridlines on the chart (dotted, solid, dashed), their colour and their width.
- **Markers:** It provides options to change the shape, size and colour of markers.
- **Series labels:** It has options to change the series label position, Values (Font, Size, Colour) and background color of the series labels.



2. General

It has the following options:

- **Properties:** We can change the size (height and width) of the visual, the position (horizontal, vertical) of the visual, and the padding of the visual.
- **Title:** We can change the title; the colour, font, font size, font style (bold, italic, underline) of the title; the subtitle; the divider and the spacing.
- **Effects:** This is very useful. It has the following options:
 - **Background:** It is used to change the background color of the visual and the transparency
 - **Visual border:** It is used to add a border to the selected visual. We can change the colour of the border and the roundedness of the corners.
 - **Shadow:** We can add a shadow to the visual. The colour, offset and position of the shadow can be customized.
- **Header icons:** It is used to change the header icons like drill on dropdown, drill up, drill down, Pin, Focus mode, Filter etc. We can choose which icons to display and we can also change the colour of the icons.
- **Tooltips:** Tooltips provide contextual information and more details about data points in a visual. They can include many things like text, images and visuals and appear when you hover over the data points on your visual. We can change the text colour and the background of the tooltips using the options in the Format Visual tab.
- **Alt text:** We can enter an alt text, i.e. a description that will be read by a screen reader on selecting the visual.



3. Add further analyses to your visual

This can be used to add extra elements to our visual. We can add a trend line, a constant x-axis line, a constant y-axis line, min/max line, average/median/percentile line and error bars among other options.

B. Format Page

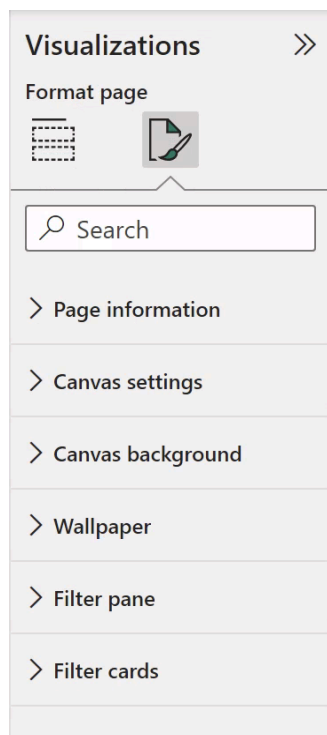


Image 3

The format page tab provides a bunch of options to make cosmetic changes to the page. These are explained below:

- **Page information:** We can change page information like page name, use page as tooltip etc.

- **Canvas settings:** We can change the type of canvas (16:9, 4:3, letter, custom etc.) and the vertical alignment.
- **Canvas background:** We can change the colour and transparency of the canvas background and can also add an image to the background.
- **Wallpaper:** We can change the colour and transparency of the wallpaper and can add an image as a wallpaper.
- **Filter pane:** We can change the text (Font, Size, Colour), input (input box colour), header (header text size), search box (search text size), border (colour) and background (colour, transparency).
- **Filter cards:** Within the filter pane are cards - filters on this page and filters on all pages. We can change the text (Font, Size, Colour), input (input box colour), border (colour) and background (colour, transparency).

2. Drill through in Power BI

In the build visual tab of the visualisation pane, there is a drill through section. It has the following options:

- **Cross report:** If we enable cross report, we can jump from a visual on one page to another page in the Power BI Desktop. We can use Cross report drill through to connect two or more visuals.

For this, in the Power BI Settings, the cross report drill through option must be enabled. These steps can be followed for the same:

- a. Go to File → Options and settings → Options.
- b. In the Options window, at the bottom of the Current File section, select Report settings.
- c. Under Cross-report drillthrough, select Allow visuals in this report to use drillthrough targets from other reports.
- d. Click on the OK button.

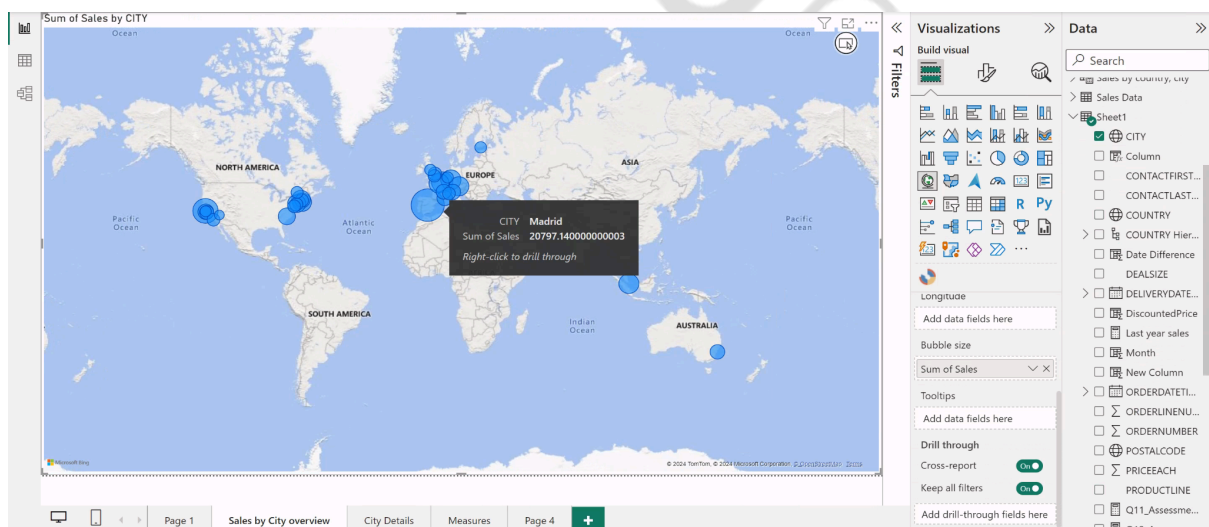
We must have a 'drill through to' page (e.g. City Details) and a main page (e.g. Sales by City Overview) from where we will drill through on a particular visual. To set up a drill through, follow the below steps:

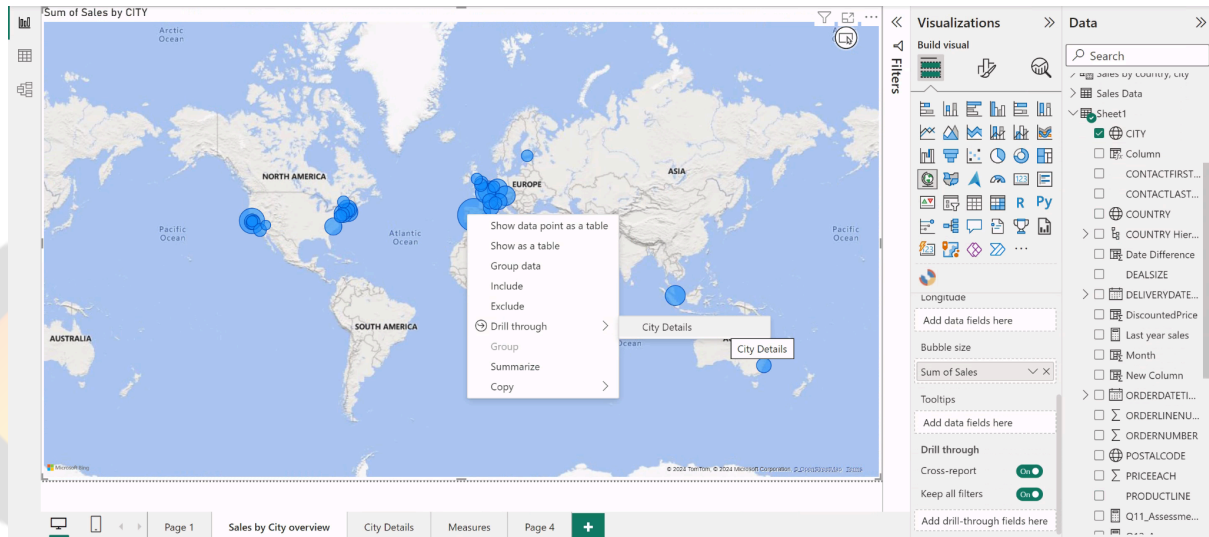
- a. Go to the 'drill through to' page (e.g. City Details).
- b. In the drill through section in the visualisation pane add a drill through field (e.g. city).

CITY	Classic Cars	Motorcycles	Planes	Ships	Trains	Trucks and Buses	Vintage Cars	Total
Aarhus	59,461.20			1,753.06	6,964.53	7,300.00	2,000.00	77,478.79
Allentown	22,634.66	26,529.02	8,100.00		5,351.61	24,961.90	8,801.08	96,378.27
Barcelona	4,500.00	9,898.90	14,984.64	27,200.66			17,342.80	73,927.00
Bergamo	5,107.22		63,558.42	10,521.04			47,956.96	127,143.64
Bergen	46,254.81	25,561.52	2,900.00				11,622.78	86,339.11
Boras	29,395.68	12,388.60	7,435.88	23,074.62		23,775.52	19,819.59	115,889.89
Boston	30,555.27	13,224.06		30,201.76	7,344.75	27,813.77	24,782.93	133,922.54
Brickhaven	61,115.25	15,709.27	17,718.98	12,826.08			33,775.96	141,145.54
Bridgewater	37,934.28	27,285.73	9,352.01		4,449.39			79,021.41
Brisbane	5,146.51					6,731.65	27,636.18	39,514.34
Bruxelles	3,000.00			26,490.72	6,555.41		32,522.05	68,568.18
Burbank	3,500.00		7,120.96			4,600.00	26,285.27	41,506.23
Burlingame	9,900.00	31,369.34	32,337.59			22,852.82	6,029.38	102,489.13
Cambridge	20,993.81	36,132.52	7,699.92	3,838.00		26,693.06	18,009.45	113,366.76
Charleroi	11,700.00		4,100.00	2,312.24	1,711.26		6,137.20	25,960.70
Chatswood	32,722.57	2,780.90	21,248.38	1,089.36		14,837.51	60,544.66	133,223.38
Cowes	3,300.00		22,626.74	25,346.55	4,598.40		17,298.44	73,170.13
Dublin	19,057.82	4,696.32	10,436.10		3,112.60	3,700.00	2,234.40	43,237.24
Espoo	58,747.72	12,584.73			5,117.05	12,303.76		88,753.26
Frankfurt	41,813.08				5,043.42	8,100.00	11,988.87	66,945.37
Genève	93,344.91							93,344.91
Glen Waverly	10,714.30		3,843.84	3,070.40			38,511.08	56,139.62
Glendale	6,400.00	6,889.40	34,214.88	4,207.66			9,611.29	61,323.23
Graz	18,758.07					18,062.18	9,684.95	46,505.20
Helsinki	30,415.36	25,034.18	20,028.02	6,407.92			7,313.67	89,199.15
København	53,882.26		7,000.00	33,077.61	3,800.00		17,508.97	115,268.84
Koln	71,544.63	7,497.50						79,042.13
Las Vegas	53,297.67						19,531.38	72,829.05
Lille	6,391.92	50,464.47						56,856.39
Liverpool	53,502.42	32,055.91					7,692.26	93,250.59
London	21,647.35		13,949.18	23,920.09		25,280.40	25,112.35	109,909.37
Los Angeles	23,918.90		5,680.15	5,727.43			6,767.30	42,093.78
Luleå	20,981.94			6,440.00	3,200.00	15,786.92	11,965.35	58,374.21
Total	2,968,546.40	971,086.29	877,942.21	677,940.40	203,804.26	947,355.18	1,644,212.05	8,290,886.79

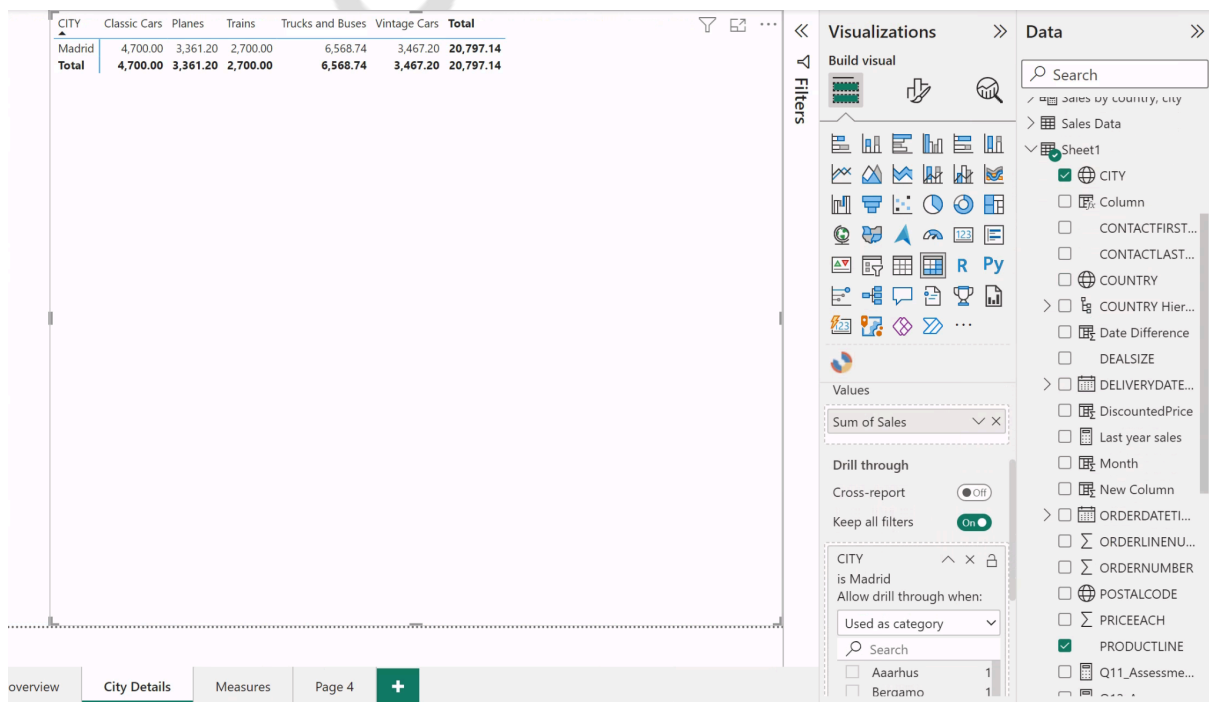
- Go to the main page from where you will drill through (e.g. Sales by City Overview).
- Right click on the field you want to drill through by (e.g. city).

Note: Make sure this field is present in the drill-through filters on the 'drill through to' page (where you will finally land on drilling through this visual).





e. You will reach the ‘drill through to’ page and the page will be filtered for the value (e.g. city).



In the example above (from [our dataset](#)), we drilled through a visual which contained the **sum of sales by city**. We jumped to a visual which contained the **sum of sales for each product line in that city**. We drilled through the City.

- **Keep all filters:** While drilling through, if we want to pass the filters in the original (source) visual to the new page (target visual), we need to keep the ‘Keep all filters’ toggle on. If you don’t want to pass the filter context while drilling through from a visual on one page to that on another page, then keep this toggle off.

3. Bookmarks in Power BI

Bookmarks can be used to capture the current state of a report page. We can save the current filters, slicers, sort order, drill location etc. Others can save their own bookmarks or use the ones created by you. Bookmarks are very useful as we can save multiple bookmarks in any order and use them to create a story because every bookmark conveys a different insight.

Steps to create a bookmark:

- Create the view that you want to save by adding visuals, and filtering data.
- Go to the View tab on the ribbon in the report view.
- Select the Bookmarks option.
- The bookmarks pane appears. In this pane, click on the Add button to add a new bookmark.
- You can perform multiple operations on this bookmark like update, rename, delete etc.

To display a bookmark, simply select it from the Bookmarks pane.

We can also choose which properties are applied by each bookmark in the report. In other words, we can limit the functionality of the bookmark based on which properties we want to see. To make these changes, select More options (...) next to the bookmark name in the Bookmark pane. Then select the property: Data, Display, and other controls.

The options are:

- Data properties, such as filters and slicers.
- Display properties, such as spotlight and its visibility.
- Current page changes, which present the page that was visible when the bookmark was added.
- We can also choose between All visuals and Selected visuals.

For example, sometimes we don't want the filters to be reset when the users are switching among different views using bookmarks. In that case, we can uncheck the Data property in each bookmark.

You can arrange bookmarks in a particular order and use the View tab in the bookmarks pane to create a slide show.

Also, we can create bookmark groups. For example, we can create a group for each country's sales manager. That person would be able to see the bookmarked view for each of his/her regional sales managers. For creating a group, simply select the Group option in the More options for each bookmark.

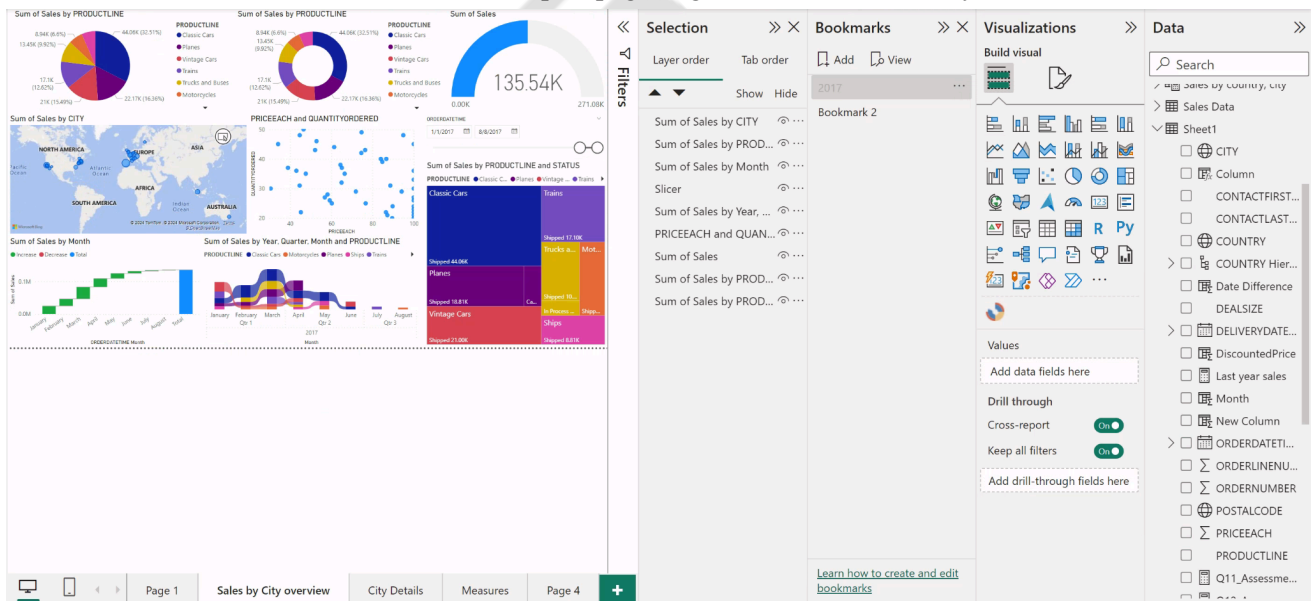
In our dataset, for example, we can create a new bookmark (USA Classic Cars - 2016) for saving all the previously created visualisations for USA for 2016 for the Classic Cars product line.



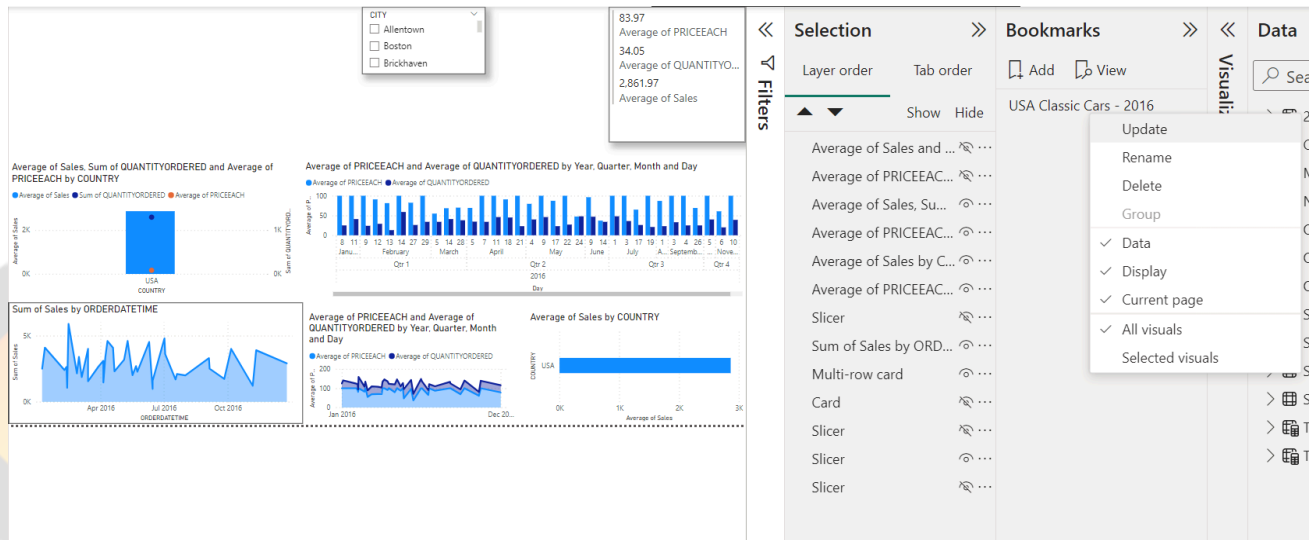
Selection pane

The selection pane helps us select which visual we want to display. We can specify which visuals should be visible by selecting the eye icon next to each visual in the selection pane. Bookmarks also save the visibility status of visuals.

Note that slicers continue to filter a report page, regardless of whether they're visible or not.



For example, if we want to update the bookmark created (USA Classic Cars - 2016) to include some particular selections, then we can simply make the selections (visuals that are required to be displayed), right click on the bookmark and select the **Update** option. Now the bookmark will include the filtered data for classic cars in the USA for 2016 and also will display some selections only, not all of them.



Utility of bookmarks

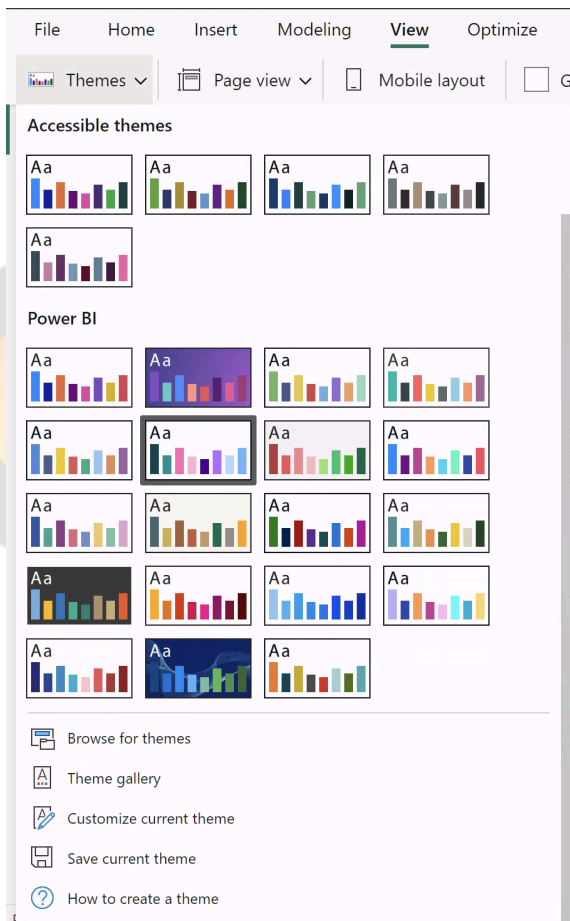
Example 1: Suppose we have a **store inventory and sales dashboard** and we have multiple views - one for each store. Each store view has different filters and visualisations. Instead of creating multiple pages, we can simply save each store view as a bookmark. For store 1, let's suppose we just need to see the **total inventory**, **total sales by value** and **number of customers visiting each day**. For store 2, which turns out to be a superstore/a larger store, we intend to track many more metrics like **avg order value**, **most ordered items**, and **shortage of inventory** along with the previous metrics. For each store we can create a different view and save it as a bookmark. Each bookmark saves the corresponding selections and filters accordingly.

Further, in the above example, we can group bookmarks as per regions (North, South, East, West) / territories / states etc.

Example 2: We can also create bookmarks to **track our progress** of creating reports. If we are creating a complex/lengthy report then we can capture the status of our report using a bookmark. It saves the filters, visuals and selections that we have made.

4. Themes in Power BI

A theme is a standardised template with set colours, backgrounds and formatting applied to your visuals. To apply a theme go to View → Themes → select the theme you want to apply.

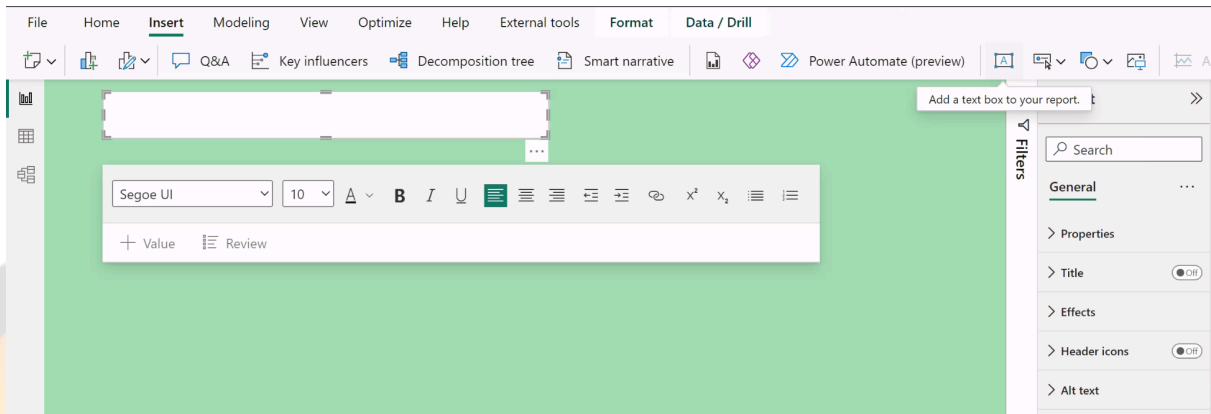


- We can select out of the available options: different kinds of predefined color schemes.
- You can browse for themes on your PC or add new themes from the online Theme gallery (<https://community.fabric.microsoft.com/t5/Themes-Gallery/bd-p/ThemesGallery>).
- We can customise the currently selected theme. We can adjust a current theme and then save it as a custom theme.
- We can also create a new theme altogether and then save it as a new custom theme. For example: this may be required when we need to use corporate colours, define new formatting and use new icon sets.

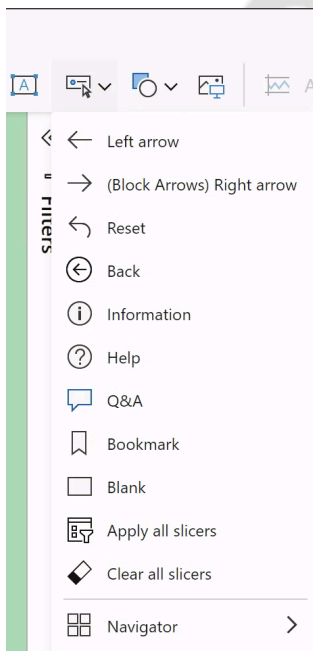
5. Adding other elements in Power BI

We can add other elements to the report to make it visually more appealing. These elements also help us build a cohesive narrative and a more powerful story around the data. Power BI provides us with these elements:

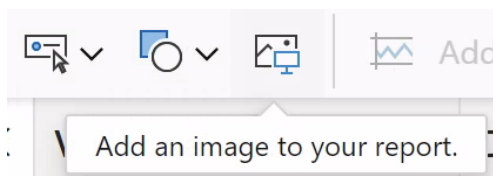
- **Textbox:** Can be added to display text anywhere in the report. Power BI also provides many formatting options for formatting the text in the report.



- **Buttons:** We can add buttons such as the left arrow, back, information in Power BI.



- **Images:** We can also add images to our report.



- **Shapes:** We can also add shapes in Power BI. The available options are shown below:

